

Midterm 1 Practice 1

INSTRUCTIONS:

- **You have three hours to complete this exam. Exceeding the three hour limit will result in a penalized score.**
 - Please read each question below carefully, noting that the exam is printed double-sided. You should write your answers on the exam itself. If you require extra space, you may write on the back page of the exam or use scratch paper and staple it to the exam.
 - There are 100 total points on the exam, 40 points from 20 questions in Part 1, and 60 points from 3 questions in Part 2.
 - You must show your work in Part 2 for full credit; showing work for Part 1 is not required (unless otherwise indicated), though may lead to partial credit.
 - You may use a testing center issued calculator.
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Part 1: Short Answer (40 points)

1. Select **all** of the following statements about the GDP Deflator and the CPI which are true:

- (a) The GDP Deflator measures prices over a fixed basket of items, whereas the CPI allows that basket to change over time.
 - (b) The CPI measures prices for both goods and services, whereas the GDP Deflator only measures the prices of goods.
 - (c) The CPI includes imported foreign goods, whereas the GDP Deflator only includes domestic goods.
 - (d) The CPI doesn't account for improvements in the quality of items purchased, whereas the GDP Deflator does.
- (c)

2. Explain what we mean by forward guidance, as it pertains to monetary policy, and why it may be useful for keeping the economy stable.

Forward guidance is the practice of the Federal Reserve providing information about expected future monetary policy actions. This practice may be beneficial to the economy by helping anchor public expectations about future inflation.

3. Why is it that the unemployment rate can be an imperfect measure of labor market conditions?

The unemployment rate can be an imperfect measure of labor market conditions because it does not account for workers outside the labor force who may be discouraged and desiring to work, but not searching. Because of the way the unemployment rate is constructed, if unemployed persons leave the labor force because of discouragement, the unemployment rate will go down.

4. True/False. One reason economists are concerned about declining population growth rates in high income countries is because slowing growth in the population may lead to slowing innovation, with fewer people entering the world to generate new ideas.

True.

5. Consider our model of national income, where $\bar{Y} = C(\bar{Y} - \bar{T}) + I(r) + \bar{G}$. Provide a graph of the supply and demand for loanable funds in this economy, and illustrate how an increase in government spending affects the equilibrium interest rate.

See Figure 3-9 in the Textbook.

6. Name two reasons for the recent slowdown in China's growth in GDP per capita.

The reasons we discussed in class were that policy initiatives boosting China's GDP per capita were not sustainable, a long period of high economic growth may have led to some accumulated inefficiencies in some sectors of the economy, recent recessions have hurt demand for China's exported products, China may be approaching its steady state, and China has recently been pivoting to more consumption-led growth (i.e., reducing savings rates).

7. According to the reading and the discussion in class, what justification was given for why is it reasonable to measure growth in economic well-being using income per worker?

This is because income per worker is correlated with many other things we care about. Alternatively, we also know that this is a statistic easily computed for many countries, whereas many other indicators of well-being are more difficult to consistently measure.

8. Select **all** of the following reasons which we gave in class as primary reasons for why there could be structural unemployment:

- (a) Laid-off workers need time to update their resumes before they are able to begin looking for work again.
 - (b) Unions may be able to pressure employers to pay a wage to union members that is higher than the going rate in non-unionized workplaces.
 - (c) The government passes a minimum wage well above what the wage would otherwise be in equilibrium.
 - (d) Because high-skilled workers are trying to find jobs that are a good match to their skills, they only apply to jobs which pay higher wages.
- (b), (c)

9. True/False. One of the three fundamentals of modern macro models is that the model should be consistent with data evidence on GDP, prices and employment.

False.

10. Explain the two core factors described in class that account for elevated inflation in the last few years.

Supply – COVID-19 disrupted supply chains and the workforce, constraining the amount of goods and services that could be purchased.

Demand – recent policy actions have boosted demand (i.e., increased stimulus both through monetary and fiscal channels), putting upward pressure on prices.

11. Suppose that demand for real money balances is given by $\frac{M}{P} = (2Y)/(i^{0.5})$, where Y is real GDP and i is the nominal interest rate. If $Y = 775$ and $i = 4$, what will velocity be?

Velocity is defined as $V = \frac{PY}{M}$. Using the demand for real money balances, we get $\frac{M}{PY} = \frac{2}{i^{0.5}}$ or $V = \frac{i^{0.5}}{2} = 1$.

12. What is the unemployment insurance program?

A social program funded by taxes on employers (and employees) that supplements a certain fraction of unemployed workers' incomes (30-50% usually) for a particular period of time (usually up to 26 weeks) while they search for a new job.

13. Select **all** of the following statements about the labor share, Cobb-Douglas production, and U.S. inequality which are **true**:

- (a) If production is Cobb-Douglas and markets are competitive, then an increase in the amount of capital in the economy will have no impact on the labor share.
 - (b) Since the pandemic, wage growth has been fastest for lower income workers (10th percentile).
 - (c) Under perfect competition and constant returns to scale, our model of national income predicts that real wages should be proportional to the average productivity of workers.
 - (d) One reason we gave in class for why the labor share may be declining is that the rise of automation, with capital displacing workers (and their income).
- (a), (b), (c), (d)
14. Explain the difference between exogenous and endogenous variables.
Exogenous variables are those variables known before hand and fed into the model and endogenous variables are the variables of interest that come out of a model.
15. Explain why we can think of seigniorage as an inflation tax.
When the government increases revenue by printing money, the increase in the money supply may lead to increased inflation, according to the quantity theory of money. Thus, the value of money held by individuals in the economy falls, while the government gains revenue, which is like if the government were to tax individuals holding money to gain revenue.
16. Which **one** of the following statements about the U.S. money supply and monetary policy is **false**?
- (a) Since the middle of 2022, the U.S. money supply has been declining.
 - (b) During the pandemic, the Federal Reserve changed banking regulations which generated an artificial surge in the measured amount of M1.
 - (c) The monetary base measures the sum of all currency and deposits in the economy.
 - (d) The Federal Reserve commonly adjust the supply of money in the economy by buying or selling assets like short-term government debt.
- (c)
17. True/False. In the Solow growth model and in the data, countries with higher savings rates had faster growth in GDP per worker.
False. Countries with higher savings rates had higher *levels* of GDP per worker, not growth rates.
18. Explain, using words and math, what the marginal propensity to consume is.
The marginal propensity to consume is the amount of extra (aggregate) consumption that occurs with an extra amount of disposable income. Mathematically, this looks like $\frac{\partial C(Y-T)}{\partial (Y-T)}$.

19. Give a definition of investment, consistent with how we have used the word in class.
Goods purchased by individuals or business for future use (i.e., machines, buildings, etc.).
20. True/False. According to the classical dichotomy, changes in real variables, like consumption or investment, will have no impact on nominal variables, such as inflation.
False. The classical dichotomy says the reverse – nominal variables will not impact real variables.

Part 2: Long Answer (60 points)

1. Consumption, Workers and the Golden Rule in 1860s Massachusetts (20 points):

Consider the economy of 1860s Massachusetts, which is described well by the Solow Growth Model. Production of goods and services is given by $Y_t = K_t^\alpha L_t^{1-\alpha}$. Output can only be used for consumption or investment, where investment $I_t = sY_t$, and consumption $C_t = (1-s)Y_t$. Capital accumulates according to the equation, $K_{t+1} = (1-\delta)K_t + I_t$. Assume that there is no population growth or technological change. Further, $\alpha = 0.3$, $s = 0.3$, and $\delta = 0.07$.

- (a) Solve for k_{gold}^* , the level of capital per worker in the steady state that generates the maximum steady state consumption per worker. Also, solve for the savings rate needed to reach that steady state level of capital per worker. (9 points)

To solve for k_{gold}^* , you maximize consumption as follows:

$$\begin{aligned} \max_{k^*} c^* &= (1-s)f(k^*) \\ &= f(k^*) - sf(k^*) \\ &= f(k^*) - \delta k^* \end{aligned}$$

where this last step follows from the statement that in equilibrium, $sf(k^*) = \delta k^*$. Note – you needn't show all work to get the per worker variables here and what these relationships are, but it will be difficult to set this problem up if you don't do that first (or know the formulas).

Now, taking a derivative with respect to k^* and setting it equal to zero, we get:

$$\begin{aligned} f'(k_{gold}^*) - \delta &= 0 \\ \alpha(k_{gold}^*)^{\alpha-1} &= \delta \\ 0.3(k_{gold}^*)^{0.3-1} &= 0.07 \\ 4.29 &= (k_{gold}^*)^{0.7} \\ 7.996 &= k_{gold}^* \end{aligned}$$

Finally, to solve for s_{gold} , we take the expression for the steady state level of capital per worker, $sf(k^*) = \delta k^*$, and plug in k_{gold}^* and solve for s_{gold} :

$$\begin{aligned} s_{gold}f(k_{gold}^*) &= \delta k_{gold}^* \\ s_{gold}(7.996)^{0.3} &= 0.07(7.996) \\ s_{gold} &= 0.3 \end{aligned}$$

Note: it is always true in our set up that $s_{gold} = \alpha$. But not showing the work for that result and just stating that this is the case will not get full points here.

- (b) In the 1860s, the population of Massachusetts is affected by the American Civil War, as many men go off to war and do not return, leading to a one-time permanent decline in the number of workers, L . Assume that the economy of Massachusetts starts in the golden rule steady state. First, draw a graph that illustrates the steady state of the Solow Growth model. Second, show on that graph how the change in L affects capital per worker. Be sure to label carefully the x-axis, all lines and all intersection points. Finally, answer the following questions: What will be the immediate impact of the decline in workers on income per worker, y_t ? Describe how y_t will grow in the years that follow and how, in the long run, it will compare to the original steady state value of y_t before the war. (8 points)

The answer is similar to the graph in the solutions to Problem Set 5 number 2b. Because there are fewer workers, $k \equiv \frac{K}{L}$ rises, leading to a higher income initially. However, because this is above the steady state level of capital per worker, capital per worker will fall and so will income per worker, having a negative growth rate. Ultimately, however, it will return to the same steady state as before. This perhaps goes without saying, but this simple exercise is unlikely to capture all the costs of war on an economy.

- (c) The March family, with four daughters Meg, Jo, Beth, and Amy, live in 1860s Massachusetts, and while income is important to them, they care more about consumption per worker, c_t . With the loss of all the workers in the economy, these “Little Women” are concerned about how fast consumption will grow now.

Based on your answer in part (b), what do you predict will happen to consumption as a result of the reduction of population during the war? You do not need to use any equations in your answer in this question unless it is helpful to you. (3 points)

The decrease in the population increases capital per worker, as seen in part (b), which will increase income per worker. Since there is no change in the savings rate, this will increase consumption as well. However, since this increase in capital per worker (and thus income per worker) is temporary, consumption per worker will decline back to its original steady state value as capital per worker declines. That is, the behavior of consumption per worker over time will qualitatively look the same as income per worker.

2. **Unemployment, Worker Flows and the Labor Force in a World with “Supers” (20 points):** Suppose the United States economy has a labor market where the job finding rate per month (f) is 0.28 and the job separation rate (s) is 0.02 and unemployment and employment evolve according to the following equations:

$$\begin{aligned}U_{t+1} &= (1 - f)U_t + sE_t \\E_{t+1} &= (1 - s)E_t + fU_t\end{aligned}$$

- (a) For a long time “Supers” have been stigmatized in the United States, forced into hiding because of lawsuits and public relations fiascos. However, the tides are changing and some former “Supers” are going from being outside the labor force to being employed again. Suppose that the U.S. economy starts at $t = 0$ with an unemployment rate of 0.067 and that the total size of the labor force is initially 15 million. With the “Supers” returning, both the labor force and employment increase by 1 million (in $t = 1$). Thus, employment is 1 million people larger than its steady state value at $t = 1$ while unemployment is initially unchanged. Using this information and the equations and number for part (a), solve for the unemployment rate at $t = 1$ and $t = 2$. (10 points)

The labor force is initially 15 million people and with an unemployment rate of 0.067, this implies that there are 1 million unemployed people ($0.067 * 15$). Thus, there are 14 million employed people. In $t = 1$, there are now 15 million employed people, still 1 million unemployed people and a labor force of 16 million. Thus, at $t = 1$, we have $\frac{U_1}{L_1} = \frac{1 \text{ million}}{16 \text{ million}} = 0.0625$. In $t = 2$, we need to use the evolution equation to figure out the number of unemployed people:

$$\begin{aligned}U_2 &= (1 - f)U_1 + sE_1 \\&= (1 - 0.28)1 + 0.02(15) \\&= 1.02\end{aligned}$$

There are now 1.02 million people, a labor force of 16 million and thus the unemployment rate is given by:

$$\frac{U_2}{L_2} = \frac{1.02}{16} = 0.06375$$

- (b) With Elastigirl being a prime example of former “Supers” returning to the labor force, two of her children, Violet and Dash, are trying to better understand better how labor markets work. Violet proposes that the standard model of unemployment flows is insufficient to understand recent labor market patterns, as many “Supers” were previously outside the labor force. As a result, Violet proposes the following simple modification of the unemployment flows model:

$$\begin{aligned}NL_{t+1} &= (1 - j)NL_t + qE_t \\U_{t+1} &= (1 - f)U_t + sE_t \\E_{t+1} &= (1 - s - q)E_t + fU_t + jNL_t\end{aligned}$$

where NL_t is the number of people outside the labor force at time t , j is the fraction who choose to join the labor force each time period, and q is the fraction of employed who leave the labor force each period. That is, the size of the labor force can now change over time. Note – to keep the model simple, Violet has assumed that people entering or exiting the labor force do not pass through unemployment.

Derive what the steady state unemployment rate is in Violet’s model. (10 points)
While this problem may seem daunting initially, it is actually fairly straightforward. Converting the NL equation into the steady state, we get:

$$\begin{aligned}\bar{NL} &= (1 - j)\bar{NL} + q\bar{E} \\j\bar{NL} &= q\bar{E}\end{aligned}$$

Now, take the employment equation and you get:

$$\bar{E} = (1 - s - q)\bar{E} + f\bar{U} + j\bar{NL}$$

Using what we know from the first equation, we get that:

$$\begin{aligned}\bar{E} &= (1 - s - q)\bar{E} + f\bar{U} + q\bar{E} \\&= (1 - s)\bar{E} + f\bar{U}\end{aligned}$$

This puts us right back in the original model, except that we’re working with the employment equation. But take the same step of setting $\bar{E} = \bar{L} - \bar{U}$ and you get:

$$\begin{aligned}\bar{L} - \bar{U} &= (1 - s)(\bar{L} - \bar{U}) + f\bar{U} \\s(\bar{L} - \bar{U}) &= f\bar{U} \\s\bar{L} &= (s + f)\bar{U} \\\frac{\bar{U}}{\bar{L}} &= \frac{s}{s + f}\end{aligned}$$

The unemployment rate in this economy is exactly the same as in the baseline economy.

3. **Production, Incomes and Profits in an Uncharted Galaxy (20 points):** Captain Christopher Pike and his crew aboard the U.S.S. Enterprise are engaged in a five-year mission to explore strange new worlds and new civilizations. As part of their mission, they encounter the planet Gamma Epsilon IV. In the economy of Gamma Epsilon IV, GDP, Y , is produced according to the production function:

$$Y = F(K, L) = 5K^{0.3}L^{0.7}$$

where K is capital and L is labor, measured as the number of workers. Assume that firms on Gamma Epsilon IV behave competitively, maximize profits, sell their product Y for nominal price P and pay a nominal wage, W , and a nominal rental rate for capital, R .

- (a) If all capital on the planet is being used, with $K = 1000$, and the nominal wage is $W = 5$ and the level of prices is $P = 2$, solve for the number of workers firms will hire (6 points).

Firms will maximize profits, and the profit function for the firm is given by:

$$Profit_N = PY - WL - RK$$

Taking the derivative with respect to labor and setting it equal to zero, we get:

$$\begin{aligned} P \frac{\partial Y}{\partial L} - W &= 0 \\ \frac{\partial F(K, L)}{\partial L} &= \frac{W}{P} \\ MPL &= \frac{W}{P} \end{aligned}$$

Given the production function, the MPL will be given by:

$$MPL = 5(0.7)K^{0.3}L^{-0.3}$$

Now all that's left is to plug in numbers and solve for L :

$$\begin{aligned} 3.5(1000)^{0.3}L^{-0.3} &= \frac{5}{2} \\ 11.121 &= L^{0.3} \\ L &= 3069.682 \end{aligned}$$

- (b) The next planet the Enterprise crew encounters is Theta Alpha III, and they observe that the production function on Theta Alpha III is given by:

$$Y = F(K, L) = AK^\alpha L^{1-\alpha}$$

Assume that there are firms on Theta Alpha III who maximize profits in the same way as the firms on Gamma Epsilon IV, albeit with different prices, different amounts of capital and labor, and different production parameters.

Unfortunately, Mr. Spock's scanner is malfunctioning, and as a result, the Enterprise crew can't directly observe the values for α , K or A . However, they do observe that $Y = 150$, $L = 200$, $MPL = 0.5$, and $MPK = 0.5$. What must be the values of α and A ? Show all work. HINTS: First, it will be helpful to think about how the labor share of income ($\frac{WL}{PY}$) relates to α and the steps of that proof. Once you know α , you will want to use what you know about the capital share ($\frac{RK}{PY}$) and α next. (7 points)

The key observation here is that $1 - \alpha = \frac{WL}{PY}$ and $\frac{W}{P} = MPL$. You could remember this or derive this; either way, you have all you need for α . Applying this, you get:

$$1 - \alpha = \frac{MPL * L}{Y} = \frac{0.5(200)}{150} = \frac{100}{150} = \frac{2}{3}$$

Thus, $\alpha = \frac{1}{3}$. Now, we still don't know K . But we do know that $\alpha = \frac{RK}{PY} = \frac{MPK * K}{Y}$. Applying this with what we know, we get:

$$\begin{aligned} \frac{1}{3} &= \frac{0.5(K)}{150} \\ K &= 100 \end{aligned}$$

Finally, now that we know K we can solve for A in a number of ways. The easiest way is to just use the production function:

$$\begin{aligned} Y &= AK^\alpha L^{1-\alpha} \\ 150 &= A(100)^{1/3}(200)^{2/3} \\ A &= 0.945 \end{aligned}$$

- (c) Finally, the crew of the Enterprise observe what is happening on the planet Beta Xi VII. For this planet, they look at the whole economy and see that it can be described by the following equations:

$$\begin{aligned}
 Y &= K^{0.5}L^{0.5} \\
 Y &= C(Y - T) + I(r) + G \\
 C(Y - T) &= 10 + 0.5(Y - T) \\
 I(r) &= 100 - 2r \\
 G &= 10 \\
 T &= 10
 \end{aligned}$$

Assume that $K = 169$ and $L = 225$. Solve for investment and the real interest rate on Beta Xi VII. (7 points)

First, solve for Y given what you know for K and L and you get:

$$\begin{aligned}
 Y &= (169)^{0.5}(225)^{0.5} \\
 &= 13(15) \\
 &= 195
 \end{aligned}$$

Then, simply plug everything in to the equilibrium equation and you get:

$$\begin{aligned}
 Y &= C(Y - T) + I(r) + G \\
 195 &= 10 + 0.5(195 - 10) + 100 - 2r + 10 \\
 195 &= 95 + 0.5(185) - 2r \\
 195 &= 212.5 - 2r \\
 8.75 &= r
 \end{aligned}$$

And thus, $I(r) = 100 - 2r = 100 + 17.5 = 82.5$.

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